DevOps Shack

Linux Troubleshooting Guide

150 Common Errors — Root Cause Analysis & Solutions

2026 Edition

Comprehensive DevOps Reference Manual for Linux Engineers

Classification: Internal Use Only

Maintained by: DevOps Shack Engineering Team

Table of Contents

How to Use This Guide

This guide covers 150 of the most common Linux errors encountered in production DevOps environments. Each entry is structured to help you quickly diagnose and resolve issues.

- ERROR: The exact error message or error code you will see in logs or terminal output.
- DESCRIPTION: What the error means in plain terms.
- ROOT CAUSE: The technical mechanism that triggers this error.
- CAUSE: The real-world scenario that leads to this error.
- SOLUTION: Step-by-step remediation actions ordered from safest/quickest to more drastic.

DESCRIPTION (Light Blue)	**ROOT CAUSE (Soft Blue)**
CAUSE (Warm Yellow)	**SOLUTION (Soft Green)**

Section 1: Filesystem & Disk Errors (1–10)

Diagram: Linux Filesystem Hierarchy & Error Points

```
┌──────────────────────────────────────────────────────────────────┐
│                  LINUX FILESYSTEM STRUCTURE                      │
└──────────────────────────────────────────────────────────────────┘
                           │
                  ┌─────────────────┐
                  │  / (root fs)    │◄── Read-only remount on error
                  └─────────────────┘
                           │
           ┌───────────────┼──────────────┐
     ┌─────────┐   ┌───────────┐   ┌─────────┐
     │  /home  │   │   /var    │   │  /tmp   │◄── Disk full / inode exhausted
     └─────────┘   └───────────┘   └─────────┘
          │              │
  Quota Limits   Log Rotation ──► Journal / Log overflow
          │
     ┌──────────────────────────────────┐
     │  STORAGE BACKEND                 │
     │  HDD/SSD ──► I/O Error (EIO)     │
     │  SMART ──► Pre-failure warning   │
     └──────────────────────────────────┘
```

#001	**No space left on device**
DESCRIPTION	The system cannot write to the disk because all available space has been consumed.
ROOT CAUSE	The filesystem reached 100% utilization, preventing any further writes including temp files, logs, or application data.
CAUSE	Uncontrolled log growth, large temp files, or a runaway process filling the disk.
SOLUTION	• Run df -h to identify the full partition • Use du -sh /* to find large directories • Remove old logs: journalctl --vacuum-time=7d • Clear package cache: apt clean or yum clean all • Archive or delete large unused files

#002	**Read-only file system**
DESCRIPTION	Attempts to write to the filesystem fail because it has been remounted read-only.
ROOT CAUSE	The kernel detected filesystem corruption or disk errors and remounted it read-only to prevent further damage.
CAUSE	Disk I/O errors, sudden power loss, or filesystem corruption.
SOLUTION	• Check dmesg for hardware errors • Run fsck /dev/sdXN after unmounting

	• Check /etc/fstab for errors=remount-ro option • Replace failing hardware if SMART errors exist: smartctl -a /dev/sdX

#003 inode: No space left

DESCRIPTION	The filesystem has run out of inodes even though disk space may remain.
ROOT CAUSE	Each file consumes one inode; millions of tiny files can exhaust the inode table before disk bytes are consumed.
CAUSE	Mail spools, session files, or cache directories accumulating millions of small files.
SOLUTION	• Check with df -i to see inode usage • Find directories with many files: find /tmp -maxdepth 1 -type d | xargs -I{} sh -c "echo {} $(ls {} | wc -l)" • Clean mail queues, tmp dirs, or cache stores • Consider reformatting with more inodes if persistent

#004 Permission denied

DESCRIPTION	A user or process attempted to access a file or directory without the required permissions.
ROOT CAUSE	UNIX permission model denied the action because the process UID/GID does not match the required permissions.
CAUSE	Incorrect file permissions, wrong ownership, or SELinux/AppArmor policy denial.
SOLUTION	• Check permissions: ls -la <file> • Fix ownership: chown user:group <file> • Fix permissions: chmod 644 <file> or chmod 755 <dir> • Check SELinux denials: audit2why < /var/log/audit/audit.log • Use sudo if elevated privileges are needed

#005 Too many open files

DESCRIPTION	A process or the system has reached the limit of simultaneously open file descriptors.
ROOT CAUSE	The OS enforces limits on open file descriptors per process (ulimit) and system-wide. Reaching these limits causes failures.
CAUSE	File descriptor leak in application code, or limits too low for high-concurrency workloads.
SOLUTION	• Check limits: ulimit -n and cat /proc/sys/fs/file-max • Increase per-process limit: ulimit -n 65536 • Set persistently in /etc/security/limits.conf • Increase system limit: sysctl -w fs.file-max=200000 • Check leaks with lsof -p <PID> | wc -l

#006 Disk quota exceeded

DESCRIPTION	A user or group has reached their assigned disk quota and cannot write more data.
ROOT CAUSE	The quota subsystem tracks disk usage per user/group and enforces hard limits set by the administrator.
CAUSE	User data growth exceeding configured quota limits.
SOLUTION	• Check quota: quota -vs <username> • View all quotas: repquota -a • Increase quota: edquota -u <username> • Help user clean up: ncdu <home_dir> • Disable quota temporarily if urgent: quotaoff /partition

#007 File not found (ENOENT)

DESCRIPTION	A process attempted to access a file or directory that does not exist at the given path.
ROOT CAUSE	The VFS layer could not resolve the path to an existing inode in the filesystem.
CAUSE	Wrong path, deleted file, unmounted filesystem, or broken symlink.
SOLUTION	• Verify path: ls -la <path> • Check symlinks: readlink -f <path> • Ensure filesystem is mounted: mount | grep <mount_point> • Check if file was deleted while in use: lsof | grep deleted

#008 Input/output error

DESCRIPTION	A low-level I/O error occurred when reading from or writing to a device.
ROOT CAUSE	The storage hardware reported an error, causing the kernel to fail the I/O request.
CAUSE	Failing disk, bad sectors, faulty cables, or RAID degradation.
SOLUTION	• Check SMART data: smartctl -H /dev/sdX • Review kernel errors: dmesg | grep -i "error\\|ata\\|sda" • Run badblocks: badblocks -sv /dev/sdX (read-only) • Replace drive if errors persist • Check cable connections and controller

#009 Filesystem corruption (fsck)

DESCRIPTION	The filesystem has structural inconsistencies detected by fsck or during mount.
ROOT CAUSE	Power loss during writes, kernel bug, or hardware failure left the filesystem in an inconsistent state.
CAUSE	Abrupt shutdown without sync, bad blocks, or journaling failure.
SOLUTION	• Boot to recovery mode

	• Unmount the partition • Run: fsck -y /dev/sdXN • Review fsck output for repaired issues • Restore from backup if fsck cannot repair

#010 Mount: wrong filesystem type

DESCRIPTION	The mount command failed because the specified filesystem type does not match the device.
ROOT CAUSE	The kernel driver for the filesystem type could not interpret the device's on-disk format.
CAUSE	Wrong -t type specified, partition not formatted, or filesystem not supported by kernel.
SOLUTION	• Auto-detect type: mount /dev/sdXN /mnt • Inspect filesystem: file -s /dev/sdXN • Load kernel module: modprobe <fstype> • Format if new disk: mkfs.ext4 /dev/sdXN

Section 2: Process & Memory Errors (11–20)

Diagram: Process Lifecycle & Memory States

```
┌──────────────────────────────────────────────────────────────┐
│                 LINUX PROCESS STATE DIAGRAM                  │
└──────────────────────────────────────────────────────────────┘

  fork()         exec()       schedule()      exit()
──────▶   CREATED ──▶ READY ──────▶ RUNNING ──────▶ ZOMBIE
                        ▲              │               │
                        │   I/O wait   ▼           wait() by
                        │        SLEEPING            parent
                        │              │               ▼
                        └──────────────┘            REAPED

Memory Zones:
┌────────────┬───────────────────┬──────────────┐
│ RAM (RSS)  │ Swap (si/so)      │ OOM Killed   │
│ < limit    │ Used when RAM     │ When both    │
│            │ exhausted         │ exhausted    │
└────────────┴───────────────────┴──────────────┘
```

#011	**Cannot allocate memory (ENOMEM)**
DESCRIPTION	The kernel refused a memory allocation request because available memory is exhausted.
ROOT CAUSE	Physical RAM and swap space are both fully committed, leaving no headroom for additional allocations.
CAUSE	Memory leak in application, insufficient swap, or overcommit limits reached.
SOLUTION	• Check memory: free -h and vmstat -s • Identify top consumers: ps aux --sort=-%mem | head • Kill leaking process: kill -9 <PID> • Add swap: fallocate -l 4G /swapfile && mkswap /swapfile && swapon /swapfile • Tune vm.overcommit_memory in /etc/sysctl.conf

#012	**Segmentation fault (SIGSEGV)**
DESCRIPTION	A process accessed memory outside its allowed address space and was terminated by the kernel.
ROOT CAUSE	The CPU's MMU triggered a protection fault when the process attempted to read/write an unmapped or protected page.
CAUSE	Bug in application code: null pointer dereference, buffer overflow, or use-after-free.

SOLUTION	• Enable core dump: ulimit -c unlimited • Analyze with gdb: gdb <binary> core • Run under valgrind for memory errors • Check for updates or patches to the software • Review recent code changes

#013 OOM Killer: process killed

DESCRIPTION	The Out-Of-Memory killer terminated a process to reclaim memory and prevent system panic.
ROOT CAUSE	The kernel OOM killer activates when memory is critically low and selects a process to kill based on OOM score.
CAUSE	Memory overcommit, no swap, or a runaway memory-consuming process.
SOLUTION	• Review: dmesg | grep -i "oom" • Identify which process was killed and why • Protect critical services: echo -17 > /proc/<PID>/oom_adj • Add more RAM or swap • Fix memory leaks in application

#014 Process not found (no such process)

DESCRIPTION	An attempt to signal or interact with a PID that does not exist in the process table.
ROOT CAUSE	The target process has already exited, was never started, or the PID was recycled.
CAUSE	Race condition, script using stale PID file, or process crashed before signal was sent.
SOLUTION	• Verify process: ps aux | grep <name> • Check PID files for staleness • Use process name instead: pkill -f <name> • Implement proper PID file management in scripts

#015 Resource temporarily unavailable (EAGAIN)

DESCRIPTION	A non-blocking I/O operation or resource acquisition failed because the resource is temporarily busy.
ROOT CAUSE	The kernel returned EAGAIN to indicate the caller should retry later, as the resource is not immediately available.
CAUSE	File lock contention, pipe full, or socket buffer full in non-blocking mode.
SOLUTION	• Implement retry logic with exponential backoff • Check file locks: lsof | grep LOCK • Increase kernel socket buffer: sysctl -w net.core.rmem_max=26214400 • Review application's non-blocking I/O handling

#016 Killed (signal 9)

DESCRIPTION	A process received SIGKILL and was immediately terminated without cleanup.
ROOT CAUSE	SIGKILL cannot be caught or ignored; it directly instructs the kernel to destroy the process.
CAUSE	OOM killer, manual kill -9, or watchdog timeout.
SOLUTION	• Determine why kill -9 was needed vs graceful kill -15 • Check OOM events in dmesg • Investigate why graceful shutdown failed • Ensure application handles SIGTERM properly

#017 Zombie process

DESCRIPTION	A dead process remains in the process table because its parent has not yet called wait() to collect its exit status.
ROOT CAUSE	The kernel retains process table entries for dead processes until the parent reads the exit status.
CAUSE	Bug in parent process not handling SIGCHLD, or parent is also stuck.
SOLUTION	• Identify zombies: ps aux | grep Z • Find parent: ps -o ppid= -p <zombie_pid> • Send SIGCHLD to parent: kill -CHLD <parent_pid> • Kill parent if it does not handle children properly • Fix application to call wait() on child exit

#018 CPU soft lockup detected

DESCRIPTION	The kernel watchdog detected that a CPU has been spinning in kernel code for more than 20 seconds.
ROOT CAUSE	A kernel thread or interrupt handler held a lock or spun in a tight loop, starving other work on that CPU.
CAUSE	Kernel bug, driver issue, or misconfigured interrupt affinity.
SOLUTION	• Review: dmesg | grep "soft lockup" • Update kernel to latest stable version • Check for known driver bugs • Reboot to clear (data may be at risk) • File bug with kernel version and lockup trace

#019 Load average too high

DESCRIPTION	System load average significantly exceeds the number of CPU cores, indicating saturation.
ROOT CAUSE	Load average counts processes in R (running) and D (uninterruptible sleep) state; high values mean queuing.

CAUSE	CPU-bound workloads, I/O wait from slow storage, or process count explosion.
SOLUTION	• Inspect: top or htop • Check I/O wait: iostat -x 1 • Identify D-state processes: ps aux | awk "$8 == "D"" • Scale horizontally or vertically • Tune I/O scheduler: cat /sys/block/sda/queue/scheduler

#020 `ulimit: max user processes exceeded`

DESCRIPTION	A new process cannot be forked because the user has reached their maximum process/thread count.
ROOT CAUSE	The OS limits the number of processes per user via nproc to prevent fork bombs and resource exhaustion.
CAUSE	Too many threads/processes spawned by services, or nproc limit too low.
SOLUTION	• Check: ulimit -u • Increase in /etc/security/limits.conf: * soft nproc 65536 • Restart affected services after adjusting limits • Check for runaway thread-spawning in applications

Section 3: Networking Errors (21–30)

Diagram: TCP/IP Network Connection Lifecycle

```
CLIENT                         KERNEL                        SERVER
  |                              |                             |
  |—— connect() ————————————————▶|                             |
  |                          SYN |————————————————————————————▶|
  |                              |◀———————————————————— SYN-ACK
  |                          ACK |————————————————————————————▶|
  |                              |         ESTABLISHED          |
  |—— send() ——————————————————▶|——— DATA ——————————————————▶|
  |◀—— recv() ————————————————|◀——— DATA ————————————————————|
  |                              |                             |
Errors:                          |                             |
ECONNREFUSED ◀— RST ———————————| (port closed)               |
ETIMEDOUT ◀— no SYN-ACK ———————| (firewall/down)             |
ENETUNREACH ◀— no route ——————— kernel routing table          |
```

<table>
<tr><th colspan="2">#021 Connection refused (ECONNREFUSED)</th></tr>
<tr><td>DESCRIPTION</td><td>A TCP connection attempt was rejected by the remote host because no service is listening on the specified port.</td></tr>
<tr><td>ROOT CAUSE</td><td>The remote kernel sent a TCP RST because the destination port has no socket in LISTEN state.</td></tr>
<tr><td>CAUSE</td><td>Service not started, wrong port, firewall RST rule, or service crashed.</td></tr>
<tr><td>SOLUTION</td><td>• Check service status: systemctl status <service>
• Verify listening ports: ss -tlnp | grep <port>
• Check firewall: iptables -L or ufw status
• Start service: systemctl start <service>
• Verify correct host/port in client config</td></tr>
</table>

<table>
<tr><th colspan="2">#022 Connection timed out (ETIMEDOUT)</th></tr>
<tr><td>DESCRIPTION</td><td>A network connection attempt did not receive a response within the timeout period.</td></tr>
<tr><td>ROOT CAUSE</td><td>TCP SYN packets were sent but no SYN-ACK was received, causing the connection to expire after retransmit attempts.</td></tr>
<tr><td>CAUSE</td><td>Firewall silently dropping packets, incorrect routing, host down, or network congestion.</td></tr>
<tr><td>SOLUTION</td><td>• Test reachability: ping <host>
• Trace route: traceroute <host>
• Check firewall rules on both ends
• Verify routing table: ip route
• Use tcpdump to confirm SYN packets leaving interface</td></tr>
</table>

#023 Network is unreachable	
DESCRIPTION	The kernel has no route for the destination network in the routing table.
ROOT CAUSE	When no route matches the destination, the kernel returns ENETUNREACH to the application.
CAUSE	Missing default gateway, downed network interface, or misconfigured routing table.
SOLUTION	• Check routes: ip route show • Add default gateway: ip route add default via <gateway_ip> • Bring up interface: ip link set eth0 up • Verify network config files in /etc/netplan/ or /etc/network/interfaces

#024 DNS resolution failure	
DESCRIPTION	A hostname cannot be resolved to an IP address, blocking network connections.
ROOT CAUSE	The resolver library (glibc) could not obtain a valid DNS answer from any configured nameserver.
CAUSE	Wrong DNS server in /etc/resolv.conf, DNS server down, NXDOMAIN, or network outage.
SOLUTION	• Test: dig @8.8.8.8 <hostname> • Check /etc/resolv.conf for valid nameservers • Verify DNS service: systemctl status systemd-resolved • Flush DNS cache: resolvectl flush-caches • Check /etc/nsswitch.conf for hosts order

#025 SSH: Connection reset by peer	
DESCRIPTION	An established SSH connection was unexpectedly terminated by the remote side.
ROOT CAUSE	The TCP connection was closed with RST rather than FIN, indicating an abrupt termination.
CAUSE	SSH server crash, firewall idle timeout, network interruption, or fail2ban blocking.
SOLUTION	• Check sshd status: systemctl status sshd • Review /var/log/auth.log for errors • Set ClientAliveInterval in sshd_config • Check fail2ban: fail2ban-client status sshd • Verify firewall idle timeout settings

#026 Address already in use (EADDRINUSE)	
DESCRIPTION	A service cannot bind to a network address/port because another process already owns it.
ROOT CAUSE	TCP/UDP socket bind fails if the exact address:port combination is already in use.

CAUSE	Previous instance of service not fully stopped, TIME_WAIT state, or port conflict.
SOLUTION	• Find owner: ss -tlnp | grep <port> or lsof -i :<port> • Kill conflicting process • Set SO_REUSEADDR in application • Wait for TIME_WAIT to expire (typically 60s) • Change service port if conflict is persistent

#027 iptables: no chain/target/match

DESCRIPTION	An iptables command references a chain, target, or match module that does not exist.
ROOT CAUSE	iptables requires specific kernel modules to be loaded for each match extension or custom chain to exist.
CAUSE	Typo in chain name, kernel module not loaded, or missing iptables extension.
SOLUTION	• List chains: iptables -L -n • Load module: modprobe xt_recent • Check module availability: lsmod | grep xt_ • Verify command syntax in iptables man page

#028 Broken pipe (EPIPE)

DESCRIPTION	A write to a pipe or socket failed because the reading end has been closed.
ROOT CAUSE	The kernel delivers SIGPIPE (or returns EPIPE) when a process writes to a pipe with no readers.
CAUSE	Pipeline commands where a downstream process exits before upstream finishes writing.
SOLUTION	• Ignore SIGPIPE in long-running services: signal(SIGPIPE, SIG_IGN) • Check that downstream pipeline commands are running • Handle EPIPE return code gracefully in code • Review log for which process exited early

#029 Network interface not found

DESCRIPTION	A network management command references an interface name that does not exist.
ROOT CAUSE	The kernel's network subsystem has no registered interface with the given name.
CAUSE	Interface renamed by udev, driver not loaded, or VM NIC not attached.
SOLUTION	• List interfaces: ip link show • Check driver: lspci | grep Ethernet then modprobe <driver> • Check udev naming rules in /etc/udev/rules.d/ • For VMs, verify NIC attached in hypervisor

#030 SSL certificate verify failed

DESCRIPTION	A TLS/SSL handshake failed because the server's certificate could not be verified against trusted CAs.
ROOT CAUSE	The TLS library could not build a valid chain from the server cert to a trusted root CA in the local trust store.
CAUSE	Expired certificate, self-signed cert not trusted, wrong hostname, or outdated CA bundle.
SOLUTION	• Check cert: openssl s_client -connect host:443 • Update CA bundle: update-ca-certificates • Add custom CA: cp ca.crt /usr/local/share/ca-certificates/ && update-ca-certificates • Check system date is correct: timedatectl • Verify certificate hostname matches SNI

Section 4: Authentication & User Errors (31–40)

Diagram: Linux Authentication Flow (PAM)

```
┌─────────┐   ┌──────────────┐   ┌──────────────────────────┐
│  User   │   │  PAM Stack   │   │  Authentication Backends │
└────┬────┘   └──────┬───────┘   └──────────┬───────────────┘
     │               │                       │
 login/ssh           │                       │
     ├──────────────►│ pam_unix.so ─────────► /etc/shadow
     │               │ pam_ldap.so ─────────► LDAP server
     │               │ pam_krb5.so ─────────► Kerberos KDC
     │               │ pam_faillock.so ─────► Lock counter
     │               │ pam_securetty.so ────► TTY check
     │◄──────────────┤
   SUCCESS         FAILURE
     │               │
  Session        auth.log
  Opened         /var/log
```

#031 Authentication failure

DESCRIPTION	A login attempt failed because the provided credentials do not match stored records.
ROOT CAUSE	PAM compared the provided password hash against /etc/shadow and found no match, denying access.
CAUSE	Wrong password, account locked, expired password, or PAM misconfiguration.
SOLUTION	• Reset password: passwd <username> • Check account status: chage -l <username> • Unlock account: usermod -U <username> • Review PAM config: /etc/pam.d/sshd • Check /var/log/auth.log for details

#032 sudo: user not in sudoers

DESCRIPTION	A user attempted to run sudo but is not authorized in the sudoers configuration.
ROOT CAUSE	sudo checks /etc/sudoers and drop-in files; if the user is not listed, the command is denied and logged.
CAUSE	User was not added to sudo group or sudoers file.
SOLUTION	• Add to sudo group (as root): usermod -aG sudo <username> • Or edit sudoers safely: visudo • Add entry: username ALL=(ALL:ALL) ALL • Log off and back on for group to take effect

#033 Account is locked

DESCRIPTION	Login attempts fail because the account has been administratively locked or auto-locked after failed attempts.
ROOT CAUSE	PAM pam_tally2/pam_faillock increments failure counters and locks accounts after threshold is exceeded.
CAUSE	Too many failed login attempts, or administrator manually locked the account.
SOLUTION	• Unlock with pam_tally2: pam_tally2 --reset --user <username> • Or pam_faillock: faillock --reset --user <username> • Or usermod: usermod -U <username> • Review /var/log/auth.log for source of failures

#034 SSH: Too many authentication failures

DESCRIPTION	SSH client disconnects after trying too many keys, triggering the server's MaxAuthTries limit.
ROOT CAUSE	sshd counts each authentication method attempt; exceeding MaxAuthTries causes disconnection.
CAUSE	SSH agent has too many keys loaded, or client retries with multiple methods before password.
SOLUTION	• Specify key: ssh -i ~/.ssh/specific_key user@host • Limit agent keys: ssh-add -D then add only needed key • Use IdentitiesOnly yes in ~/.ssh/config • Increase MaxAuthTries in sshd_config (temporary)

#035 su: must be run from a terminal

DESCRIPTION	The su command was invoked without a controlling terminal, which PAM requires for interactive authentication.
ROOT CAUSE	PAM's pam_securetty module or tty check requires a proper TTY for the su authentication flow.
CAUSE	Running su inside scripts, cron jobs, or non-interactive shells without TTY allocation.
SOLUTION	• Use sudo -i instead of su • Allocate PTY in SSH: ssh -t user@host su - • Use PAM configuration that allows non-TTY su • Configure passwordless sudo for the specific command needed

#036 Group does not exist

DESCRIPTION	A command references a group name that is not present in /etc/group.
ROOT CAUSE	The group database (/etc/group and optionally LDAP/NIS) has no entry matching the given group name.

CAUSE	Typo in group name, group was deleted, or LDAP connectivity issue.
SOLUTION	• List groups: getent group | grep <groupname> • Create group: groupadd <groupname> • Check LDAP connectivity if using directory services • Verify nsswitch.conf group lookup order

#037 User does not exist

DESCRIPTION	A command references a username that has no entry in the user database.
ROOT CAUSE	The user database (/etc/passwd or directory service) returns no result for the given username.
CAUSE	User not created, typo, deleted account, or LDAP unreachable.
SOLUTION	• Check: id <username> or getent passwd <username> • Create user: useradd -m <username> • Check LDAP/SSSD connectivity • Verify /etc/nsswitch.conf passwd entry

#038 PAM: system error

DESCRIPTION	A PAM module encountered an unexpected system-level error during authentication.
ROOT CAUSE	A PAM module failed due to missing library, misconfiguration, or system call failure.
CAUSE	Corrupted PAM config, missing PAM module file, or broken library dependencies.
SOLUTION	• Review /etc/pam.d/ for syntax errors • Check module exists: ls /lib/security/pam_*.so • Run: pamtester <service> <user> authenticate • Review /var/log/auth.log for specific error • Reinstall pam package if module is missing

#039 Kerberos: kinit failed

DESCRIPTION	Kerberos ticket initialization failed, preventing SSO authentication.
ROOT CAUSE	The KDC rejected the authentication request or the client configuration is incorrect.
CAUSE	Wrong password, clock skew >5 minutes, unreachable KDC, or wrong realm.
SOLUTION	• Check time sync: timedatectl and chronyc tracking • Verify KDC reachability: nc -zv <kdc_host> 88 • Check /etc/krb5.conf realm and KDC settings • Verify username and realm spelling

#040 LDAP: Can't contact LDAP server

DESCRIPTION	The LDAP client cannot establish a connection to the configured directory server.
ROOT CAUSE	The TCP connection to the LDAP port (389 or 636) failed or timed out.
CAUSE	LDAP server down, wrong URI, firewall blocking, or TLS misconfiguration.
SOLUTION	• Test: ldapsearch -x -H ldap://<server> -b dc=example,dc=com • Check connectivity: nc -zv <ldap_server> 389 • Verify /etc/ldap/ldap.conf or sssd.conf settings • Check TLS certs if using LDAPS • Restart sssd: systemctl restart sssd

Section 5: Systemd & Service Errors (41–50)

Diagram: systemd Service Start Flow

```
systemctl start <service>
          |
          ▼
    ┌──────────────────────────────────────────┐
    │  1. Resolve unit file                    │
    │  2. Check dependencies (Requires/After)  │◄── Dependency failed?
    │  3. Apply resource limits (cgroup)       │
    │  4. Execute ExecStartPre=                │
    │  5. Execute ExecStart=                   │◄── Binary not found?
    │  6. Wait for readiness (Type=notify)     │◄── Timeout?
    │  7. Execute ExecStartPost=               │
    └──────────────────────────────────────────┘
                   |
                   |
      ┌──────────────────────────┐
      │ active (running)         │   or   failed → journalctl -u <svc>
      └──────────────────────────┘
```

#041 systemd: Unit failed to start

DESCRIPTION	A systemd service unit entered the failed state and could not start.
ROOT CAUSE	The service's main process exited with a non-zero code or could not be executed.
CAUSE	Missing executable, configuration error, dependency not met, or resource limit hit.
SOLUTION	• Check status: systemctl status <service> • View logs: journalctl -u <service> -n 50 • Test config: <binary> --configtest • Check dependencies: systemctl list-dependencies <service> • Run ExecStart manually to see errors

#042 Failed to enable unit: File exists

DESCRIPTION	systemctl enable failed because the symlink target already exists.
ROOT CAUSE	An existing file or symlink at the expected symlink path conflicts with the enable operation.
CAUSE	Leftover symlink from previous install, or unit enabled through alternate method.
SOLUTION	• Check: ls -la /etc/systemd/system/multi-user.target.wants/ | grep <unit> • Remove stale symlink manually • Use systemctl --force enable <unit> • Run systemctl daemon-reload after fixing

#043 Job for service failed because a timeout was exceeded

DESCRIPTION	systemd killed a service start/stop operation because it did not complete within the configured timeout.
ROOT CAUSE	systemd sends SIGTERM then SIGKILL if a service does not transition state within TimeoutStartSec or TimeoutStopSec.
CAUSE	Service hung during initialization, slow disk, network dependency, or inadequate timeout.
SOLUTION	• Increase timeout in unit file: TimeoutStartSec=120 • Check why service hangs: strace -p <PID> • Review service logs: journalctl -u <service> • Ensure network dependencies are met before start

#044 systemd: Dependency failed

DESCRIPTION	A service failed to start because one of its required dependencies could not start.
ROOT CAUSE	systemd's dependency graph propagates failures; if a Requires= unit fails, dependents are also failed.
CAUSE	A dependent service (database, network, mount) failed to start.
SOLUTION	• Check: systemctl list-dependencies --failed <service> • Fix the failing dependency first • Use After= instead of Requires= for optional deps • Check journalctl -u <dependency> for root cause

#045 cgroup: Resource limit exceeded

DESCRIPTION	A process in a cgroup was killed or throttled because it exceeded the cgroup resource limit.
ROOT CAUSE	cgroup v1/v2 enforces memory, CPU, and I/O limits; violations result in OOM kill or throttling.
CAUSE	MemoryMax or MemoryLimit in systemd unit too restrictive for workload.
SOLUTION	• Check limit: systemctl show <service> | grep Memory • Adjust in unit file: MemoryMax=2G • Override without editing unit: systemctl edit <service> • Monitor: systemd-cgtop

#046 Service: ExecStart not found

DESCRIPTION	systemd cannot start a service because the specified executable does not exist.
ROOT CAUSE	The ExecStart= path in the unit file does not resolve to an existing executable.
CAUSE	Software not installed, wrong path in unit file, or binary removed.
SOLUTION	• Check path: which <binary> or ls -la <path> • Install missing package • Correct ExecStart= path in unit file

	• Run systemctl daemon-reload after editing unit

#047 journald: No space left for journal

DESCRIPTION	systemd-journald cannot write new log entries because the journal disk quota is reached.
ROOT CAUSE	journald enforces SystemMaxUse and SystemKeepFree limits on journal storage.
CAUSE	Excessive logging, long retention, or misconfigured journal size limits.
SOLUTION	• Check journal size: journalctl --disk-usage • Vacuum by time: journalctl --vacuum-time=7d • Vacuum by size: journalctl --vacuum-size=1G • Adjust /etc/systemd/journald.conf: SystemMaxUse=500M

#048 Failed to mount tmpfs

DESCRIPTION	A tmpfs (RAM-backed) filesystem could not be mounted, preventing /tmp or similar from initializing.
ROOT CAUSE	The kernel could not allocate the requested size from available memory for the tmpfs mount.
CAUSE	Insufficient free memory, or size= option exceeds available RAM+swap.
SOLUTION	• Check available memory: free -h • Reduce tmpfs size: mount -o remount,size=512M /tmp • Add swap if memory is insufficient • Review /etc/fstab tmpfs size options

#049 Reached target shutdown but service still running

DESCRIPTION	During shutdown, one or more services did not stop cleanly within the timeout.
ROOT CAUSE	systemd sends SIGTERM during shutdown and waits DefaultTimeoutStopSec; unresponsive services get SIGKILL.
CAUSE	Service not handling SIGTERM, deadlock, or waiting on I/O.
SOLUTION	• Identify stuck service: journalctl -b -1 | grep "Stopping\\\|Timeout" • Fix SIGTERM handling in application • Reduce TimeoutStopSec in unit file • Pre-stop hook to gracefully drain connections

#050 Timer unit missed activation

DESCRIPTION	A systemd timer unit could not fire at its scheduled time.
ROOT CAUSE	If the system was off or the timer service failed when the scheduled time passed, the activation is missed.

CAUSE	System was powered off, or timer service was disabled.
SOLUTION	• Enable persistent timers: Persistent=true in timer unit • Check timer status: systemctl list-timers • Verify: systemctl is-enabled <timer>.timer • Review: journalctl -u <timer>.timer

Section 6: Package Management Errors (51–60)

Diagram: Package Manager Resolution Flow (apt/dnf)

```
  User: apt install <package>
                |
                ▼
  ┌──────────────────┐     miss      ┌─────────────────────┐
  | Local DB Cache   |──────────────▶| Repository Server   |
  └──────────────────┘               └─────────────────────┘
                | hit                        | package + checksum
                ▼                            ▼
  ┌──────────────────┐              ┌─────────────────────┐
  | Dependency Solve |              | GPG Verification    |◀── Key missing?
  └──────────────────┘              └─────────────────────┘
                | all deps ok                | valid
                ▼                            ▼
  ┌──────────────────┐              ┌─────────────────────┐
  |   dpkg/rpm       |◀─────────────|   Download .deb/    |
  |   Install        |              |   .rpm file         |
  └──────────────────┘              └─────────────────────┘
```

#051 dpkg: dependency problems prevent configuration

DESCRIPTION	A Debian package cannot be configured because its dependencies are not satisfied.
ROOT CAUSE	dpkg's post-installation configuration scripts require dependencies to be fully installed first.
CAUSE	Interrupted install, manually removed dependency, or version conflict.
SOLUTION	• Fix with apt: apt-get -f install • Check broken packages: dpkg --audit • Force configure: dpkg --configure -a • Reinstall problematic package: apt reinstall <package>

#052 rpm: failed dependencies

DESCRIPTION	RPM refuses to install a package because required dependencies are not present.
ROOT CAUSE	RPM's dependency resolver requires all Requires: entries to be satisfied before installation.
CAUSE	Missing dependency, wrong architecture, or repository not enabled.
SOLUTION	• Use dnf which auto-resolves: dnf install <package> • Enable required repo: dnf config-manager --enable <repo> • Force without deps (risky): rpm --nodeps -i <package>.rpm • Check: rpm -qpR <package>.rpm for requirements

#053 apt: Unable to lock the administration directory

DESCRIPTION	The apt/dpkg lock file is held by another process, preventing concurrent package operations.
ROOT CAUSE	dpkg uses /var/lib/dpkg/lock-frontend to prevent concurrent modifications to the package database.
CAUSE	Another apt/dpkg process running (including unattended-upgrades), or stale lock.
SOLUTION	• Check for running process: ps aux | grep -E "apt|dpkg" • Wait for completion, or kill if stale • Remove stale lock (only if no apt running): rm /var/lib/dpkg/lock* • Run: dpkg --configure -a

#054 yum/dnf: No package available

DESCRIPTION	The package manager cannot find the requested package in any enabled repository.
ROOT CAUSE	No repository in the enabled set has metadata for the requested package name.
CAUSE	Typo in package name, package in disabled repo, or EPEL/third-party repo needed.
SOLUTION	• Search: dnf search <keyword> • Enable EPEL: dnf install epel-release • Check all repos: dnf repolist all • Add third-party repo if needed • Verify package name and arch

#055 pip: Could not find a version that satisfies requirement

DESCRIPTION	pip cannot find a package version matching the specified requirement in PyPI or configured indexes.
ROOT CAUSE	pip queries configured indexes and finds no release matching the version specifier and Python version.
CAUSE	Package name typo, network issue, Python version incompatibility, or private PyPI not configured.
SOLUTION	• Search: pip search <package> or check pypi.org • Specify compatible Python: pip3 install or python3.9 -m pip • Add index: pip install --index-url <url> <package> • Check network/proxy settings

#056 GPG key verification failed

DESCRIPTION	Package manager refuses to install a package because GPG signature verification failed.

ROOT CAUSE	The package's cryptographic signature does not match the signing key in the keyring, indicating possible tampering.
CAUSE	Key not imported, expired key, or package from unofficial source.
SOLUTION	• Import key: apt-key add <keyfile> or rpm --import <keyfile> • For apt: curl -fsSL <key_url> | gpg --dearmor | tee /etc/apt/keyrings/<name>.gpg • Verify key fingerprint from official source • Check key expiry: apt-key list

#057 Repository metadata outdated

DESCRIPTION	Package manager reports that cached repository metadata is stale or corrupt.
ROOT CAUSE	The local metadata cache does not match what the repository server has, causing inconsistency.
CAUSE	Network failure during metadata sync, or repository was updated since last sync.
SOLUTION	• Update cache: apt update or dnf makecache • Clean cache first: dnf clean all && dnf makecache • Check repo URLs in /etc/yum.repos.d/ or /etc/apt/sources.list

#058 snap: snap is already installed

DESCRIPTION	A snap installation fails because the snap is already present in the system.
ROOT CAUSE	snapd tracks installed snaps and rejects duplicate installs.
CAUSE	Attempting to install an already installed snap without using refresh.
SOLUTION	• Update instead: snap refresh <snapname> • Check installed: snap list • Remove first if needed: snap remove <snapname>

#059 pip: externally managed environment

DESCRIPTION	pip refuses to install packages system-wide because the Python installation is managed by the OS package manager.
ROOT CAUSE	PEP 668 requires pip to respect OS package manager ownership to prevent conflicts.
CAUSE	Trying to pip install on Python managed by apt/dnf without a virtual environment.
SOLUTION	• Use virtual environment: python3 -m venv .venv && source .venv/bin/activate • Or use --user flag: pip install --user <package> • Or use pipx for CLI tools: pipx install <tool> • Override (not recommended): pip install --break-system-packages <package>

#060 checksum mismatch during download

DESCRIPTION	The downloaded package file's checksum does not match the expected value from the repository metadata.
ROOT CAUSE	The package manager verifies SHA256 checksums against repository metadata; a mismatch indicates corruption.
CAUSE	Network corruption, CDN cache issue, or man-in-the-middle attack.
SOLUTION	• Re-download: dnf clean packages && dnf install <package> • Try different mirror: dnf --disablerepo=* --enablerepo=<specific> install <package> • Check network for packet corruption • Verify no MITM: compare with official checksum

Section 7: Docker & Container Errors (61–70)

Diagram: Docker Container Lifecycle

```
   docker run <image>
          |
          ├─► Pull image (if not local) ──────────────────► Registry
          |    (auth check ► layer download ► checksum verify)
          |
          ├─► Create container (overlayfs layers)
          |    image layers (read-only) + container layer (read-write)
          |
          ├─► Configure namespaces & cgroups
          |    pid / net / mnt / user / uts / ipc
          |
          ├─► OCI runtime (runc) start
          |
          └─► Container RUNNING
                  |          |
            PID exit     OOM Kill
                  |          |
            exit code   exit code 137
            (inspect → docker inspect <id>)
```

#061 docker: Cannot connect to the Docker daemon

DESCRIPTION	The Docker client cannot communicate with the Docker daemon socket.
ROOT CAUSE	The Docker client connects to /var/run/docker.sock; if the daemon is stopped or the socket has wrong permissions, connection fails.
CAUSE	Docker daemon not running, user not in docker group, or socket permission issue.
SOLUTION	• Check daemon: systemctl status docker • Start daemon: systemctl start docker • Add user to docker group: usermod -aG docker $USER • Log out/in for group change to take effect • Verify socket: ls -la /var/run/docker.sock

#062 docker: No space left on device

DESCRIPTION	Docker operations fail because the Docker storage backend has run out of disk space.
ROOT CAUSE	Docker stores images, containers, and volumes in /var/lib/docker; this partition can fill up with unused resources.
CAUSE	Accumulated unused images, stopped containers, or large volumes.
SOLUTION	• Check: docker system df • Prune all unused: docker system prune -a --volumes

	• Remove images: docker image prune -a • Remove containers: docker container prune • Expand /var/lib/docker partition or move to larger disk

#063 OCI runtime: permission denied

DESCRIPTION	Docker or containerd cannot start a container due to a permission error in the OCI runtime.
ROOT CAUSE	The container runtime (runc) encounters a permission denial when setting up namespaces, cgroups, or filesystem mounts.
CAUSE	SELinux/AppArmor policy, wrong user namespace configuration, or missing kernel capabilities.
SOLUTION	• Check: docker logs <container> • Inspect SELinux: ausearch -m AVC | tail • Add SELinux label: docker run --security-opt label=disable • Check if user namespaces are enabled: sysctl user.max_user_namespaces

#064 docker build: COPY failed

DESCRIPTION	A Dockerfile COPY instruction fails during build.
ROOT CAUSE	The build context does not contain the specified source path, or the destination path cannot be written.
CAUSE	.dockerignore excluding needed files, wrong source path, or insufficient disk in build layer.
SOLUTION	• Check .dockerignore is not excluding needed files • Verify source path relative to Dockerfile location • Use --no-cache to debug: docker build --no-cache • Check available disk space on build host

#065 Container exits with code 137

DESCRIPTION	A Docker container exits immediately with code 137, indicating it was killed by SIGKILL.
ROOT CAUSE	Exit code 137 = 128 + 9 (SIGKILL); the container's main process received SIGKILL from the OOM killer or Docker.
CAUSE	Container exceeded its memory limit, OOM kill, or docker stop timing out.
SOLUTION	• Check: docker inspect <container> | grep OOMKilled • Increase memory limit: docker run -m 2g • Check logs before exit: docker logs <container> • Profile memory usage and fix leaks

#066 docker pull: unauthorized authentication required

DESCRIPTION	A docker pull fails with an authorization error from the registry.
ROOT CAUSE	The registry returned HTTP 401, indicating the client is not authenticated or credentials are invalid.
CAUSE	Not logged in to registry, expired credentials, or wrong registry URL.
SOLUTION	• Login: docker login <registry> • Check credentials: cat ~/.docker/config.json • For ECR: aws ecr get-login-password | docker login • Verify image name includes correct registry hostname

#067 Kubernetes: CrashLoopBackOff

DESCRIPTION	A Kubernetes pod repeatedly crashes and is restarted by kubelet in an exponential backoff loop.
ROOT CAUSE	The container's main process exits with a non-zero code, causing kubelet to restart it with increasing delays.
CAUSE	Application error, missing environment variable, failed dependency, or OOM kill.
SOLUTION	• Check logs: kubectl logs <pod> --previous • Describe pod: kubectl describe pod <pod> • Check resource limits: kubectl get pod <pod> -o yaml | grep -A5 resources • Fix application crash or adjust resource limits

#068 Kubernetes: ImagePullBackOff

DESCRIPTION	Kubernetes cannot pull the container image for a pod.
ROOT CAUSE	kubelet failed to pull the image and is in exponential backoff, retrying after increasing delays.
CAUSE	Wrong image name/tag, private registry without credentials, or registry unreachable.
SOLUTION	• Describe pod: kubectl describe pod <pod> • Create registry secret: kubectl create secret docker-registry • Reference in pod: imagePullSecrets • Verify image exists: docker pull <image> from node • Check registry DNS from cluster nodes

#069 docker network: endpoint with name already exists

DESCRIPTION	Docker network creation or container connection fails because the endpoint name is already registered.
ROOT CAUSE	Docker's network model requires unique endpoint names within a network namespace.
CAUSE	Leftover endpoint from previous container that was not properly removed.

SOLUTION	• Disconnect stale endpoint: docker network disconnect -f <network> <container> • List endpoints: docker network inspect <network> • Prune networks: docker network prune • Recreate network if needed

#070 containerd: failed to create shim

DESCRIPTION	containerd cannot start the container shim process, preventing container startup.
ROOT CAUSE	The containerd-shim binary cannot be executed due to permission, missing binary, or cgroup issues.
CAUSE	Missing shim binary, incompatible versions between containerd and runc, or SELinux denial.
SOLUTION	• Check containerd logs: journalctl -u containerd • Verify shim binary: which containerd-shim-runc-v2 • Check cgroup version compatibility • Reinstall containerd if binary is missing or corrupted

Section 8: Kernel & Boot Errors (71–80)

Diagram: Linux Boot Sequence & Error Points

```
Power On
  |
  ▼
UEFI/BIOS POST ────────────────────────────────▶ Hardware errors
  |
  ▼
GRUB Bootloader ────────────────────────────▶ filesystem/grub errors
  |  (reads /boot/grub/grub.cfg)
  ▼
Kernel Load (vmlinuz) ────────────────────▶ kernel panic
  |
  ▼
initramfs ─────────────────────────────────▶ device not found
  |  (udev, dracut/mkinitramfs)
  ▼
Mount root filesystem ────────────────────▶ wrong uuid/fstype
  |
  ▼
systemd (PID 1) ───────────────────────────▶ service failures
  |
  ▼
Login prompt (getty)
```

#071 `Kernel panic: not syncing`

DESCRIPTION	The kernel encountered an unrecoverable error and halted the system.
ROOT CAUSE	A kernel panic occurs when the kernel cannot continue safely. It dumps state and stops.
CAUSE	Corrupted kernel, missing initramfs, hardware failure, or critical driver bug.
SOLUTION	• Boot from rescue disk • Check boot logs for panic message • Reinstall kernel: dnf reinstall kernel • Rebuild initramfs: dracut --force or update-initramfs -u • Check hardware for failures: memtest86+

#072 `GRUB: error: unknown filesystem`

DESCRIPTION	GRUB fails to read the boot partition and cannot load the kernel.
ROOT CAUSE	GRUB modules attempt to read the filesystem where /boot resides; an unrecognized or corrupted filesystem causes failure.
CAUSE	Filesystem corruption, GRUB modules missing, or wrong root device specified.

SOLUTION	• Boot GRUB rescue mode • Set root: set root=(hd0,1) • Load modules: insmod ext2 • Boot: linux /boot/vmlinuz root=/dev/sda1; initrd /boot/initrd.img; boot • After boot: grub-install /dev/sda && update-grub

#073 initramfs: Unable to find a medium containing a live file system

DESCRIPTION	The initramfs cannot find and mount the root filesystem during early boot.
ROOT CAUSE	The initial ramdisk cannot resolve the root= parameter to an accessible block device.
CAUSE	Wrong root= in GRUB, UUID mismatch, missing storage driver in initramfs.
SOLUTION	• Verify UUID: blkid | grep <root_partition> • Compare with /etc/fstab UUID • Update /etc/default/grub with correct root • Rebuild initramfs with correct modules

#074 modprobe: Module not found

DESCRIPTION	The kernel module loader cannot find the specified module in the module search paths.
ROOT CAUSE	modprobe searches /lib/modules/$(uname -r)/ for the module; if absent, it fails.
CAUSE	Module not installed for current kernel, misspelled module name, or kernel upgrade without matching modules.
SOLUTION	• Verify kernel version: uname -r • Check module: find /lib/modules/$(uname -r) -name "*<module>*" • Install kernel modules: apt install linux-modules-extra-$(uname -r) • Run: depmod -a to rebuild module dependency db

#075 kernel: NMI watchdog: BUG: soft lockup

DESCRIPTION	The NMI watchdog detected a CPU stuck in kernel code for more than 20 seconds.
ROOT CAUSE	The NMI watchdog fires a hardware interrupt every 20s; if a CPU does not respond (held by spinlock or IRQ disabled), a soft lockup BUG is recorded.
CAUSE	Kernel bug, driver hanging with IRQs disabled, or extreme CPU saturation.
SOLUTION	• Check dmesg for lockup trace • Identify module from stack trace • Update kernel and affected driver • File upstream kernel bug report with trace

#076 ACPI: Warning: Unsupported BusWidth

DESCRIPTION	The kernel ACPI subsystem logs warnings about unsupported bus width in ACPI tables.
ROOT CAUSE	The ACPI firmware tables define hardware resources; inconsistencies trigger kernel warnings.
CAUSE	BIOS/UEFI firmware bug, or running newer kernel on hardware with outdated firmware.
SOLUTION	• Update BIOS/UEFI firmware • Add kernel parameter to suppress: acpi=off (test only) • Check if upstream kernel has fix for your hardware • File bug with firmware vendor

#077 dracut: Cannot find device (root)

DESCRIPTION	The dracut-based initramfs cannot locate the root block device during boot.
ROOT CAUSE	dracut's udev rules and device discovery cannot find the expected root device by UUID, label, or path.
CAUSE	UUID changed after disk replacement, missing storage driver, or corrupted udev rules.
SOLUTION	• Boot to dracut emergency shell • Manually mount root: mount /dev/sda1 /sysroot • Check /etc/fstab for correct UUID • Rebuild initramfs: dracut --force /boot/initramfs-$(uname -r).img $(uname -r)

#078 EFI: boot entry not found

DESCRIPTION	The UEFI firmware cannot find the configured boot entry and falls back to removable media or fails.
ROOT CAUSE	UEFI boot entries stored in NVRAM reference EFI applications on the ESP partition; missing/corrupt entries cause boot failure.
CAUSE	Boot entry deleted, NVRAM reset, ESP corruption, or bootloader removed.
SOLUTION	• Boot from USB recovery • List entries: efibootmgr -v • Reinstall bootloader: grub-install --target=x86_64-efi • Create entry: efibootmgr --create --disk /dev/sda --part 1 --label Linux --loader /EFI/ubuntu/grubx64.efi

#079 kernel: SCSI error: return code = 0x08000002

DESCRIPTION	The kernel SCSI subsystem reports a device error for a storage device command.

ROOT CAUSE	SCSI mid-layer received an error from the HBA or device; 0x08000002 often indicates DID_TRANSPORT_DISRUPTED.
CAUSE	Failing HBA, loose SAS/SATA cable, enclosure issue, or SAN path failure.
SOLUTION	• Check: dmesg | grep -i scsi • Run storage diagnostics: smartctl -a /dev/sdX • Check SAN/storage array event logs • Reseat cables and HBA • Enable multipath if not already configured

#080 memory: EDAC error detected

DESCRIPTION	The EDAC (Error Detection and Correction) subsystem reports correctable or uncorrectable memory errors.
ROOT CAUSE	EDAC hardware monitors ECC RAM for bit errors; single-bit errors are corrected, multi-bit are uncorrectable and can panic.
CAUSE	Failing RAM DIMM, poor seating, or power supply instability.
SOLUTION	• Check: edac-util -s 4 or /sys/bus/platform/drivers/i7core_edac/ • Run memtest86+ to identify failing DIMM slot • Replace failing DIMM • Check kernel: dmesg | grep -i edac

Section 9: Storage & RAID Errors (81–90)

Diagram: Storage Stack: From Application to Disk

```
┌──────────────────────────────────────────────────────────────┐
│  Application  (read/write syscalls)                          │
└──────────────────────────────────────────────────────────────┘
                         │
                         │
┌────────────────────────▼─────────────────────────────────────┐
│  VFS Layer  (ext4 / xfs / btrfs / zfs)                       │◄─errors
└──────────────────────────────────────────────────────────────┘
                         │
                         │
┌────────────────────────▼─────────────────────────────────────┐
│  Block Layer  (I/O scheduler, bio merging)                   │
└──────────────────────────────────────────────────────────────┘
        │                               │                   │
        │                               │                   │
┌───────▼───────┐               ┌───────▼──────┐  ┌─────────▼─────┐
│ LVM / mdRAID  │               │  Multipath   │  │  NFS / iSCSI  │
└───────────────┘               └──────────────┘  └───────────────┘
        │                               │                   │
        │                               │                   │
┌───────▼───────────────────────────────▼───────────────────▼───────┐
│                Physical Drive  (SCSI / NVMe / SAN)                │
└───────────────────────────────────────────────────────────────────┘
```

#081 LVM: Couldn't find device with uuid

DESCRIPTION	LVM cannot find a Physical Volume by its stored UUID, preventing Volume Group activation.
ROOT CAUSE	LVM scans configured devices for PV labels; if the device is missing or renamed, the VG cannot be fully assembled.
CAUSE	Disk removed, device renamed after udev rule change, or disk failure.
SOLUTION	• Scan for PVs: pvs and pvdisplay • Check device availability: lsblk • Update device path in LVM config if renamed • Remove missing PV (data loss): vgreduce --removemissing <vg>

#082 mdadm: ARRAY failed to assemble

DESCRIPTION	A software RAID array cannot be assembled because required member disks are missing or degraded.
ROOT CAUSE	mdadm requires a minimum number of disks to assemble an array; below that, it refuses to assemble to avoid data corruption.
CAUSE	Failed disk, wrong device node, or UUID mismatch in mdadm.conf.
SOLUTION	• Scan arrays: mdadm --examine --scan

	• Force assemble (degraded): mdadm --assemble --force /dev/md0 /dev/sd[bcde] • Check array status: cat /proc/mdstat • Replace failed disk and add: mdadm /dev/md0 --add /dev/sdX

#083 XFS: metadata I/O error

DESCRIPTION	XFS logged a metadata I/O error, indicating it cannot read or write filesystem structures.
ROOT CAUSE	XFS encountered a read or write error on a metadata block, which can lead to filesystem shutdown.
CAUSE	Failing disk, storage controller error, or bad block.
SOLUTION	• Check dmesg for underlying I/O errors • Run SMART check: smartctl -a /dev/sdX • Repair (unmounted): xfs_repair /dev/sdXN • Mount with nobarrier as temp workaround on good hardware

#084 ext4: journal commit I/O error

DESCRIPTION	The ext4 filesystem's journal cannot commit a transaction due to an I/O error.
ROOT CAUSE	ext4 uses a journal (journal area on disk) for crash consistency; I/O errors writing to the journal cause filesystem to enter read-only or error state.
CAUSE	Failing disk, bad sectors in journal area, or storage timeout.
SOLUTION	• Check dmesg: dmesg | grep -i ext4 • Check disk health: smartctl -a /dev/sdX • Force fsck: fsck -f /dev/sdXN • Disable journaling temporarily (test only): tune2fs -O ^has_journal /dev/sdXN

#085 NFS: stale file handle

DESCRIPTION	An operation on an NFS-mounted file returns a stale file handle error.
ROOT CAUSE	The file handle stored by the NFS client refers to an inode that no longer exists on the server (deleted or server reboot).
CAUSE	Server rebooted, export changed, or file deleted on server while client held it open.
SOLUTION	• Unmount and remount NFS share • Remount: umount -f /mnt/nfs && mount /mnt/nfs • Check NFS server: showmount -e <server> • Use soft,intr mount options for better recovery

#086 ZFS: pool is faulted

DESCRIPTION	A ZFS storage pool entered the FAULTED state and is offline.
ROOT CAUSE	ZFS fault management marks a pool faulted when too many vdevs are unavailable to maintain the redundancy level.
CAUSE	Multiple disk failures exceeding RAID-Z redundancy, or VDEV corruption.
SOLUTION	• Check: zpool status • Clear errors if hardware replaced: zpool clear <pool> • Replace disk: zpool replace <pool> <old> <new> • Import from cache: zpool import -f <pool> • Restore from backup if pool is unrecoverable

#087 LUKS: Failed to open the device

DESCRIPTION	A LUKS-encrypted device cannot be unlocked and opened.
ROOT CAUSE	cryptsetup verify the passphrase/keyfile against the LUKS header; failure indicates wrong key or corrupted header.
CAUSE	Wrong passphrase, corrupted LUKS header, keyfile missing, or slot erased.
SOLUTION	• Try all passphrases: cryptsetup open /dev/sdX cryptname • Check slots: cryptsetup luksDump /dev/sdX • Restore header from backup: cryptsetup luksHeaderRestore • Ensure keyfile is present and correct: cryptsetup --key-file=/path/to/key open

#088 iSCSI: connection closed

DESCRIPTION	An iSCSI initiator lost its connection to the target, causing storage I/O to fail.
ROOT CAUSE	The TCP connection between the initiator and target was interrupted; iSCSI session recovery depends on timers and configuration.
CAUSE	Network outage, target service restart, firewall change, or iSCSI target crash.
SOLUTION	• Restart iSCSI initiator: systemctl restart iscsid • Reconnect session: iscsiadm -m session --rescan • Check network path to iSCSI target • Review /var/log/messages for iSCSI errors • Tune recovery timers in iscsid.conf

#089 multipath: path failed

DESCRIPTION	Device Mapper Multipath reports that one or more paths to a storage device have failed.
ROOT CAUSE	multipathd monitors all paths to a LUN; a path failure triggers failover to surviving paths.
CAUSE	HBA port failure, SAN switch port down, or storage controller path issue.
SOLUTION	• Check: multipathd show paths

	• Check failed path: multipathd show topology • Reinstate: multipathd reinstate path <path> • Fix underlying HBA/SAN issue • Review /etc/multipath.conf for correct failover settings

#090 btrfs: parent transid verify failed

DESCRIPTION	btrfs detected a consistency error in its B-tree during a read operation.
ROOT CAUSE	btrfs stores generation numbers (transid) to verify tree nodes; a mismatch indicates that an older version of a node was written (write ordering violation).
CAUSE	Disk firmware write caching bug, power loss, or faulty storage controller.
SOLUTION	• Check btrfs: btrfs check /dev/sdXN (read-only first) • Repair (may lose data): btrfs check --repair /dev/sdXN • Enable metadata redundancy: convert to RAID1 for metadata • Replace storage hardware and restore from backup

Section 10: Security Errors (91–100)

Diagram: Linux Security Layers

```
┌──────────────────────────────────────────────────────────────────┐
│                       LINUX SECURITY STACK                       │
└──────────────────────────────────────────────────────────────────┘

 Application ──▶ PAM (authentication) ──▶ /etc/shadow / LDAP / Kerberos
      │
      ▼
 Syscall Interface
      │
      ▼
 ┌──────────────────────────────────────────────────────────────┐
 │  LSM (Linux Security Modules)                                │
 │   ├─ SELinux: context:type labels ──▶ AVC deny in audit.log  │
 │   ├─ AppArmor: path-based profiles ──▶ DENIED in syslog      │
 │   └─ Seccomp: syscall filter                                 │
 └──────────────────────────────────────────────────────────────┘
      │
      ▼
 Netfilter (iptables/nftables) ──▶ DROP/REJECT rules
      │
      ▼
 Audit subsystem (auditd) ──▶ /var/log/audit/audit.log
```

#091 SELinux: avc: denied

DESCRIPTION	SELinux blocked an operation because the security policy did not permit it.
ROOT CAUSE	SELinux's mandatory access control evaluated the source context, target context, and object class against the policy and returned a denial.
CAUSE	Incorrect file context, new application not covered by policy, or policy not updated after change.
SOLUTION	• Check: ausearch -m AVC -ts recent • Generate policy module: audit2allow -M mypol < /var/log/audit/audit.log • Load module: semodule -i mypol.pp • Fix file context: chcon -t <correct_type> /path/to/file • Or restore: restorecon -Rv /path/

#092 AppArmor: operation not permitted

DESCRIPTION	AppArmor denied an operation because no rule in the active profile allows it.
ROOT CAUSE	AppArmor profiles define an allow-list; any operation not listed is denied when in enforce mode.

CAUSE	Application attempting to access path/syscall not in profile, or profile in wrong mode.
SOLUTION	• Check: dmesg | grep apparmor or /var/log/syslog | grep DENIED • Set to complain mode: aa-complain /etc/apparmor.d/<profile> • Generate additions: aa-logprof • Edit profile: /etc/apparmor.d/<profile> • Reload: apparmor_parser -r /etc/apparmor.d/<profile>

#093 Firewalld: Error: ALREADY_ENABLED

DESCRIPTION	firewall-cmd reports that a rule or service is already enabled in the specified zone.
ROOT CAUSE	firewalld tracks enabled rules; attempting to add a duplicate returns an error.
CAUSE	Rule already added, or configuration applied twice by automation.
SOLUTION	• Query current rules: firewall-cmd --list-all • Check service state: firewall-cmd --query-service=<service> • Idempotent additions add: --permanent and ignore ALREADY_ENABLED • Use Ansible firewalld module with state: enabled

#094 fail2ban: Already banned

DESCRIPTION	fail2ban logs show an IP is already in the ban list but the ban command is attempted again.
ROOT CAUSE	fail2ban tracks banned IPs in its database; log parsing may trigger a second ban attempt if already present.
CAUSE	High log event rate causing duplicate matches, or database inconsistency.
SOLUTION	• Check ban status: fail2ban-client status <jail> • Check DB: fail2ban-client get <jail> banip • Unban if needed: fail2ban-client set <jail> unbanip <ip> • Review filter regex to prevent duplicate matches

#095 openssl: certificate has expired

DESCRIPTION	An SSL/TLS operation fails because the certificate's validity period has ended.
ROOT CAUSE	TLS implementations verify the notAfter field against system clock; past expiry causes handshake failure.
CAUSE	Certificate not renewed before expiry, or system clock is incorrect.
SOLUTION	• Check cert: openssl x509 -noout -dates -in cert.pem • Check system time: timedatectl • Renew with Let's Encrypt: certbot renew • Replace with new certificate • Set up monitoring/alerting for expiry

#096 SSH: Host key verification failed

DESCRIPTION	The SSH client refuses to connect because the server's host key does not match the stored known_hosts entry.
ROOT CAUSE	SSH clients store server fingerprints; a changed fingerprint could indicate a man-in-the-middle attack.
CAUSE	Server OS reinstalled (new host key), server IP reused, or actual MITM attack.
SOLUTION	• Verify change is expected (server rebuild) • Remove old entry: ssh-keygen -R <hostname> • Or: sed -i "/<hostname>/d" ~/.ssh/known_hosts • Reconnect and verify new fingerprint out-of-band

#097 GPG: No public key

DESCRIPTION	A GPG decryption or verification operation fails because the required public key is not in the keyring.
ROOT CAUSE	GPG looks up the key ID specified in the encrypted/signed data; if absent from the keyring, it cannot proceed.
CAUSE	Key not imported, wrong keyring, or key server unreachable.
SOLUTION	• Import from keyserver: gpg --keyserver keys.openpgp.org --recv-keys <keyid> • Import from file: gpg --import public.key • List keyring: gpg --list-keys • Verify key fingerprint after import

#098 auditd: backlog limit exceeded

DESCRIPTION	The kernel audit subsystem dropped audit events because the backlog queue overflowed.
ROOT CAUSE	auditd has a configurable backlog limit; if the kernel generates events faster than auditd reads them, overflow occurs.
CAUSE	Very high syscall rate, audit rules too broad, or auditd disk I/O slow.
SOLUTION	• Increase backlog: auditctl -b 8192 • Set in /etc/audit/auditd.conf: tcp_listen_queue=100 • Narrow audit rules to reduce event volume • Speed up audit log storage

#099 rkhunter: Warning: suspicious file found

DESCRIPTION	rkhunter detected a file that matches its database of known rootkits or suspicious file properties.
ROOT CAUSE	rkhunter compares file hashes, SUID bits, hidden files, and known rootkit signatures against its database.

CAUSE	Rootkit infection, false positive from legitimate software change, or outdated rkhunter database.
SOLUTION	• Review: rkhunter --report-warnings-only • Update database: rkhunter --update • Whitelist false positive in /etc/rkhunter.conf • If true positive: isolate system, restore from known-good backup • Forensic analysis with chkrootkit too

#100 `chroot: failed to run command`

DESCRIPTION	A chroot operation fails to execute a command inside the chroot environment.
ROOT CAUSE	The chroot changes the root directory; if required libraries, the dynamic linker, or the binary itself are missing inside the chroot, execution fails.
CAUSE	Incomplete chroot setup, missing /lib or /lib64, or binary not present inside chroot.
SOLUTION	• Check binary exists: ls <chroot>/usr/bin/<binary> • Copy required libs: ldd <binary> to find deps • Mount proc/sys/dev: mount --bind /dev <chroot>/dev • Or use schroot/debootstrap for complete chroot environments

Section 11: Advanced Networking (101–110)

Diagram: Advanced Network Troubleshooting Path

```
Is the issue PHYSICAL?
  ├ ip link show ──▶ state DOWN? ──▶ ip link set <if> up
  └ ethtool <if> ──▶ Link detected: no ──▶ cable/NIC issue
              |
          IP LAYER
  ├ ip addr show ──▶ IP assigned? ──▶ dhclient / static config
  └ ip route show ──▶ default route? ──▶ ip route add default via <gw>
              |
       TRANSPORT LAYER
  ├ ping <dst> ──▶ ICMP passes? ──▶ no ──▶ firewall?
  ├ traceroute ──▶ where does it stop?
  └ ss -tlnp ──▶ service listening on correct port?
              |
        APPLICATION LAYER
  ├ curl -v ──▶ SSL cert valid? ──▶ DNS resolving?
  └ tcpdump -i <if> port <n> ──▶ packets arriving?
```

#101 iptables: invalid source/destination

DESCRIPTION	An iptables rule addition fails because the IP address or CIDR notation is invalid.
ROOT CAUSE	iptables parses address specifications strictly; incorrect notation causes rule rejection.
CAUSE	Typo in IP, incorrect CIDR prefix, or using IPv6 address with iptables (use ip6tables).
SOLUTION	• Validate IP: ipcalc <ip>/<prefix> • Use ip6tables for IPv6 rules • Check CIDR: ensure prefix length matches address family

#102 tc: Cannot find device

DESCRIPTION	The tc (traffic control) command cannot find the specified network interface.
ROOT CAUSE	tc operates on kernel network interfaces by name; the specified interface does not exist.
CAUSE	Interface down, renamed, or not yet created (virtual interface).
SOLUTION	• List interfaces: ip link show • Check interface state: ip link show <iface> • Ensure virtual interface is created first (for tunnels/veth)

#103 nftables: Error: no such file or directory

DESCRIPTION	nftables ruleset loading fails because a referenced file (for sets or include) does not exist.
ROOT CAUSE	nft processes include directives; missing files cause a parse/load error.
CAUSE	Ruleset file deleted, wrong path in nft config, or first-run without baseline file.
SOLUTION	• Check nft file paths: nft -c -f /etc/nftables.conf • Create missing file with appropriate content • List current ruleset: nft list ruleset

#104 VPN: TLS handshake failed

DESCRIPTION	A VPN client (OpenVPN or similar) fails to establish a TLS connection with the server.
ROOT CAUSE	The TLS handshake process failed due to certificate, cipher, or version mismatch.
CAUSE	Expired certificate, unsupported cipher suite, TLS version mismatch, or firewall blocking.
SOLUTION	• Check VPN logs: journalctl -u openvpn • Verify certificate validity: openssl verify client.crt • Align tls-version-min settings between client and server • Ensure UDP/TCP port is accessible

#105 hostname: Name or service not known

DESCRIPTION	A hostname cannot be resolved via any configured resolution mechanism.
ROOT CAUSE	The resolver tried DNS, /etc/hosts, and configured NSS plugins; none returned a result for the hostname.
CAUSE	Hostname not in DNS, /etc/hosts, or search domain not configured.
SOLUTION	• Check /etc/hosts for static entries • Verify /etc/resolv.conf search domain • Test: dig <hostname> vs dig <hostname>.<domain> • Add to /etc/hosts as temporary fix: <ip> <hostname>

#106 socket: Too many open sockets

DESCRIPTION	A service cannot create a new socket because the per-process or system socket limit is reached.
ROOT CAUSE	Sockets consume file descriptors; hitting the fd limit also prevents new socket creation.
CAUSE	Socket leak in application, too many concurrent connections, or fd limit too low.
SOLUTION	• Check: ss -s for socket summary • Increase fd limit: ulimit -n 65536 • Check for socket leaks: lsof -p <PID> | grep socket | wc -l • Fix application to properly close sockets

#107 NetworkManager: device not managed

DESCRIPTION	NetworkManager reports a device as unmanaged and will not configure it.
ROOT CAUSE	NetworkManager has a list of unmanaged devices specified in its config; matching devices are ignored.
CAUSE	Interface explicitly unmanaged in /etc/NetworkManager/conf.d/, or keyfile missing.
SOLUTION	• Check: nmcli device status • Edit unmanaged list: /etc/NetworkManager/conf.d/99-unmanaged.conf • Set managed: nmcli device set <iface> managed yes • Restart NetworkManager: systemctl restart NetworkManager

#108 DHCP: No lease obtained

DESCRIPTION	A DHCP client could not obtain an IP address lease from a DHCP server.
ROOT CAUSE	DHCP DISCOVER was broadcast but no DHCP OFFER was received within the timeout.
CAUSE	No DHCP server on network, server pool exhausted, firewall blocking UDP 67/68, or MAC filtered.
SOLUTION	• Check: dhclient -v <interface> for verbose output • Verify DHCP server reachability and pool • Check firewall on DHCP server for UDP 67 • Verify MAC not on exclusion list • Use static IP as workaround

#109 BGP: hold timer expired

DESCRIPTION	A BGP session went down because the hold timer expired without receiving a KEEPALIVE or UPDATE.
ROOT CAUSE	BGP peers exchange KEEPALIVE messages; if none arrive within the hold time, the session is torn down.
CAUSE	Network outage, high CPU on router, authentication mismatch, or firewall blocking BGP port 179.
SOLUTION	• Check BGP session: vtysh -c "show bgp summary" • Verify TCP 179 connectivity between peers • Check authentication (MD5 password) match • Review router CPU: top in router shell • Tune keepalive/hold timers

#110 wireguard: Handshake did not complete

DESCRIPTION	WireGuard cannot establish an encrypted tunnel because the handshake timed out.

ROOT CAUSE	WireGuard requires UDP connectivity between peers and matching cryptographic keys for the handshake to succeed.
CAUSE	Wrong peer public key, incorrect endpoint IP/port, firewall blocking UDP, or NAT issue.
SOLUTION	• Check: wg show and verify keys • Test UDP reachability to peer endpoint • Check firewall allows WireGuard UDP port • Verify AllowedIPs match traffic • Enable persistent keepalive for NAT traversal: PersistentKeepalive = 25

Section 12: Programming & Build Errors (111–120)

Diagram: Compilation & Build Error Flow

```
  Source Code (.c/.cpp/.py/.go)
        |
    Preprocessing
    (#include, macros)
        |
    Compilation ──────────────────────────────► Syntax/Type errors
    (source → object file)
        |
    Assembly
    (.o files)
        |
    Linking ──────────────────────────────────► Undefined symbol
    (object + libraries ──► binary)             Missing -l flag
        |
    Runtime ──────────────────────────────────► ImportError (Python)
    (dynamic linking at load time)               ClassNotFoundException (Java)
        |                                        Cannot find module (Node)
      Execution
        |
    Program Running ──────────────────────────► Segfault / OOM
```

#111 gcc: undefined reference to symbol

DESCRIPTION	The linker cannot resolve a symbol referenced in the object files during compilation.
ROOT CAUSE	The linker searches specified libraries for symbols; if a library providing the symbol is not linked, the reference remains unresolved.
CAUSE	Missing -l flag, wrong library order, or library not installed.
SOLUTION	• Add missing library: gcc ... -lpthread -lm • Check library order (dependencies after dependents) • Find library: ldconfig -p | grep libname • Install dev package: apt install libname-dev

#112 cmake: Could NOT find package

DESCRIPTION	CMake cannot locate a required dependency package during the configuration phase.
ROOT CAUSE	CMake's find_package() searches standard and custom paths; if the package is not installed or not in the search path, it fails.
CAUSE	Package not installed, dev headers missing, or non-standard install prefix.
SOLUTION	• Install dev package: apt install lib<name>-dev

	• Set hint: cmake -D<Name>_ROOT=/opt/mylib .. • Install to standard prefix • Check CMakeLists.txt for exact package name required

#113 python: ImportError: No module named

DESCRIPTION	Python cannot import a module because it is not installed or not in the Python path.
ROOT CAUSE	Python's import system searches sys.path; if the module directory is not there, import fails.
CAUSE	Module not installed, wrong virtual environment, or PYTHONPATH not set.
SOLUTION	• Install: pip install <module> • Check venv: which python and pip list • Add to path: export PYTHONPATH=/path/to/module:$PYTHONPATH • Use full path to python: /path/to/venv/bin/python script.py

#114 java: ClassNotFoundException

DESCRIPTION	Java cannot find the specified class at runtime, causing the JVM to throw an exception.
ROOT CAUSE	The Java ClassLoader searches the classpath; if the class's JAR or directory is absent, loading fails.
CAUSE	JAR not in classpath, wrong package name, or conflicting class versions.
SOLUTION	• Set classpath: java -cp /path/to/jar:. MainClass • Or use: export CLASSPATH=/path/to/jar:$CLASSPATH • Check dependency in build tool (Maven/Gradle) • Verify class name spelling including package

#115 go build: cannot find package

DESCRIPTION	The Go toolchain cannot locate a package required for building.
ROOT CAUSE	Go modules resolve package paths via go.mod and the module proxy; missing or unreachable packages fail resolution.
CAUSE	Package not in go.mod, GOPROXY unreachable, or GOPATH issue in GOPATH mode.
SOLUTION	• Add dependency: go get <package@version> • Update modules: go mod tidy • Set proxy: GOPROXY=https://proxy.golang.org go build • Use vendor: go mod vendor && go build -mod=vendor

#116 make: No rule to make target

DESCRIPTION	GNU make cannot find a rule to build the specified target.
ROOT CAUSE	make searches the Makefile for a rule matching the target; if absent and no implicit rule applies, it fails.
CAUSE	Typo in target name, missing Makefile, or target not defined.
SOLUTION	• List available targets: make help or grep "^[a-z]" Makefile • Check spelling of target • Ensure you are in the correct directory with Makefile • Generate Makefile if using autotools: ./configure

#117 ruby: Gem not found

DESCRIPTION	A Ruby gem required by the application is not installed or not available in the load path.
ROOT CAUSE	Bundler and Ruby's require mechanism search GEM_PATH for the gem; absence causes LoadError.
CAUSE	gem not installed, wrong Ruby version, or bundle install not run.
SOLUTION	• Install gem: gem install <gemname> • With Bundler: bundle install • Check Ruby version: ruby -v and rbenv/rvm status • Use bundle exec <command> to use correct gem set

#118 node: Cannot find module

DESCRIPTION	Node.js cannot resolve and load the specified module during require/import.
ROOT CAUSE	Node's module resolution algorithm searches node_modules directories up the directory tree; missing modules cause failure.
CAUSE	npm/yarn install not run, node_modules deleted, or wrong module name.
SOLUTION	• Install dependencies: npm install or yarn • Check node_modules: ls node_modules/<module> • Reinstall: rm -rf node_modules && npm install • Verify module name in package.json dependencies

#119 bash: command not found

DESCRIPTION	The shell cannot find the specified command in any directory listed in PATH.
ROOT CAUSE	The shell searches each directory in PATH for an executable matching the command name.
CAUSE	Package not installed, PATH not including install directory, or typo.
SOLUTION	• Check PATH: echo $PATH • Find command: which <command> or type <command> • Install package providing command • Add to PATH: export PATH=$PATH:/new/dir

	• Check if command needs execution permission: chmod +x /path/to/cmd

#120 Ansible: UNREACHABLE!

DESCRIPTION	Ansible cannot connect to a managed host to execute tasks.
ROOT CAUSE	Ansible uses SSH (by default) to connect to targets; if the connection fails, all tasks are marked unreachable.
CAUSE	SSH not running on target, wrong credentials, firewall blocking, or inventory misconfiguration.
SOLUTION	• Test manually: ssh user@target_host • Check SSH service on target: systemctl status sshd • Verify inventory: ansible-inventory --list • Test ping: ansible <host> -m ping -vvv • Check become settings if sudo is needed

Section 13: Database Errors (121–130)

Diagram: Database Connection & Replication Topology

```
┌──────────────────────────────────────────────────────────────────┐
│                     DATABASE CLUSTER TOPOLOGY                      │
└──────────────────────────────────────────────────────────────────┘

Application ──▶ Connection Pool (PgBouncer/ProxySQL)
                           │
            ┌──────────────┴───────────┐
            ▼                          ▼
     PRIMARY / Master           READ Replica(s)
  (writes + reads)           (reads only)
            │                          ▲
            └──── Replication ─────────┘
         (binlog/WAL streaming)

Common Errors:
├─ Too many connections ──▶ Increase max_connections or use pooler
├─ Replication lag ──────▶ Check replica lag; scale read replicas
├─ Table locked/crashed ──▶ REPAIR TABLE / InnoDB recovery
└─ Write to replica ──────▶ Use primary for writes; check app config
```

#121 MySQL: Too many connections

DESCRIPTION	MySQL refuses new connections because the maximum connection count is reached.
ROOT CAUSE	MySQL maintains a global connection pool limited by max_connections; exceeding it returns error 1040.
CAUSE	Application connection leak, connection pool misconfigured, or max_connections too low.
SOLUTION	• Check: SHOW STATUS LIKE "Threads_connected" • Increase: SET GLOBAL max_connections = 500 • Set permanently in my.cnf: max_connections=500 • Implement connection pooling (ProxySQL, PgBouncer) • Fix connection leaks in application code

#122 PostgreSQL: FATAL: role does not exist

DESCRIPTION	A connection to PostgreSQL fails because the specified database role is not defined.
ROOT CAUSE	PostgreSQL authentication maps connection user to pg_roles; an unknown role is rejected.
CAUSE	Role not created, connecting as wrong user, or pg_hba.conf misconfiguration.

SOLUTION	• Create role: CREATE ROLE <rolename> LOGIN; • Check roles: \du in psql • Check pg_hba.conf allows connection for role • Ensure application uses correct username

#123 Redis: NOAUTH Authentication required

DESCRIPTION	A Redis command fails because the client has not authenticated with the required password.
ROOT CAUSE	Redis requirepass enforces password authentication; unauthenticated clients receive NOAUTH error.
CAUSE	Password set in redis.conf but client not sending AUTH command.
SOLUTION	• Authenticate: AUTH <password> • Configure client with password in connection string • Check redis.conf requirepass setting • Use ACL for fine-grained access control in Redis 6+

#124 MongoDB: Not primary or slave

DESCRIPTION	A write operation to a MongoDB replica set member fails because it is not the primary.
ROOT CAUSE	MongoDB replica sets enforce that writes go to the primary; secondaries return this error for writes.
CAUSE	Connecting to secondary directly, primary election in progress, or application not using replica set URI.
SOLUTION	• Use replica set connection string: mongodb://host1,host2,host3/db?replicaSet=rs0 • Check primary: rs.status() in mongo shell • Configure readPreference correctly for reads • Wait for election to complete if primary changed

#125 sqlite: database is locked

DESCRIPTION	An SQLite operation fails because another process has a write lock on the database file.
ROOT CAUSE	SQLite uses file-level locking; concurrent writes are serialized, and a writer blocks other writers.
CAUSE	Long-running transaction, application crash leaving lock, or concurrent processes using same DB file.
SOLUTION	• Set busy timeout: PRAGMA busy_timeout = 5000 • Check for processes: fuser <db_file> • Remove stale lock: rm <db_file>-journal (only if no writers) • Use WAL mode: PRAGMA journal_mode=WAL for better concurrency

#126 Elasticsearch: master_not_discovered_exception

DESCRIPTION	An Elasticsearch node cannot discover a master node, causing cluster health to be RED.
ROOT CAUSE	Elasticsearch requires a quorum of master-eligible nodes to elect a master; if quorum is not met, the cluster is inoperative.
CAUSE	Not enough master-eligible nodes online, split-brain, or network partition.
SOLUTION	• Check cluster state: GET /_cluster/health • Review discovery config: cluster.initial_master_nodes • Verify minimum_master_nodes (ES 6): (N/2)+1 • Ensure network connectivity between nodes • Restart down nodes

#127 MySQL: Table is crashed

DESCRIPTION	A MySQL table is in a crashed state and cannot be read or written.
ROOT CAUSE	The MyISAM storage engine (and sometimes InnoDB) can mark a table as crashed when it detects structural inconsistency.
CAUSE	Power failure during write, disk error, or MySQL killed during table operation.
SOLUTION	• Repair: REPAIR TABLE <table>; • Or: myisamchk -r /var/lib/mysql/<db>/<table>.MYI • For InnoDB: set innodb_force_recovery=1 to 6 incrementally • Restore from backup for severe corruption

#128 PostgreSQL: could not connect to server: No such file or directory

DESCRIPTION	The psql client cannot connect to PostgreSQL because the UNIX socket file does not exist.
ROOT CAUSE	PostgreSQL creates a UNIX socket in /var/run/postgresql/ by default; if the server is not running, no socket exists.
CAUSE	PostgreSQL service not running, or socket directory path mismatch.
SOLUTION	• Start service: systemctl start postgresql • Check socket: ls /var/run/postgresql/ • Specify host explicitly: psql -h 127.0.0.1 -p 5432 • Check service logs: journalctl -u postgresql

#129 RabbitMQ: Channel closed with error

DESCRIPTION	A RabbitMQ channel is closed by the broker with an error, interrupting message flow.

ROOT CAUSE	RabbitMQ closes channels when protocol violations occur, such as declaring a queue with wrong parameters or exceeding channel limits.
CAUSE	Queue property mismatch (durable/exclusive), channel limit exceeded, or resource alarm.
SOLUTION	• Check RabbitMQ logs: /var/log/rabbitmq/ • Verify queue declaration parameters match existing queue • Increase channel limit: channel_max in rabbitmq.conf • Check memory/disk alarms: rabbitmqctl status | grep alarms

#130 Cassandra: WriteTimeoutException

DESCRIPTION	A Cassandra write operation timed out because not enough replicas acknowledged the write within the timeout.
ROOT CAUSE	Cassandra requires a configurable number of replica acknowledgments (consistency level); if replicas are slow or down, the write times out.
CAUSE	Node down, GC pause, overloaded cluster, or consistency level too strict.
SOLUTION	• Check: nodetool status to see node health • Lower consistency level temporarily: CONSISTENCY ONE • Check GC: nodetool gcstats • Review write latency: nodetool tpstats • Scale cluster or tune compaction

Section 14: CI/CD & Automation Errors (131–140)

Diagram: CI/CD Pipeline Flow & Failure Points

```
Developer ──▶ git push ──▶ Source Repository (GitHub/GitLab)
                                  │
                           Webhook trigger
                                  │
                         ┌────────▼────────┐
                         │  CI Server       │◀── No runner available
                         │  (Jenkins/GL CI) │
                         └────────┬────────┘
          ┌──────────────┬────────┴─────────────────┐
          ▼              ▼                           ▼
     Build/Test     Static Analysis            Docker Build
   (unit tests)    (SonarQube QG)             (Dockerfile)
          │              │                           │
          └──────────────┴───────────────────────────┘
                               │
                      ┌────────▼────────┐
                      │  Artifact Store  │
                      │  (ECR/Nexus)     │
                      └────────┬────────┘
                               │
                    ┌──────────▼──────────┐
                    │  CD / Deploy         │◀── Helm/ArgoCD errors
                    │  (Kubernetes/EC2)    │
                    └─────────────────────┘
```

#131 Jenkins: Build failed: executor not available

DESCRIPTION	A Jenkins build cannot start because no executor is available to run it.
ROOT CAUSE	Jenkins queues jobs when all executors on matching agents are occupied; jobs remain pending indefinitely if no agent becomes free.
CAUSE	All agents busy, agent offline, or label mismatch between job and available agents.
SOLUTION	• Check agent status in Jenkins UI > Manage Nodes • Restart offline agents • Add more agents or increase executor count • Review job label requirements • Use cloud agents (EC2, Kubernetes) for elastic scaling

#132 GitLab CI: Runner not available

DESCRIPTION	A GitLab CI pipeline job is stuck because no registered runner matches the job requirements.

ROOT CAUSE	GitLab CI dispatches jobs to runners matching the job's tags and executor type; no available match leaves the job pending.
CAUSE	All runners busy, runner offline, or job tags not matching any runner.
SOLUTION	• Check runner status: Settings > CI/CD > Runners • Register new runner: gitlab-runner register • Ensure runner tags match job tags or use untagged runner • Restart runner: gitlab-runner restart

#133 Terraform: Error acquiring state lock

DESCRIPTION	Terraform cannot acquire the state lock because another operation holds it.
ROOT CAUSE	Terraform uses state locking (DynamoDB for S3 backend) to prevent concurrent modifications; an existing lock blocks new operations.
CAUSE	Previous Terraform run crashed without releasing lock, or concurrent apply.
SOLUTION	• Check lock info: terraform force-unlock <lock-id> • Verify no concurrent run is active • Check DynamoDB table for stale lock entry • Use -lock=false only in emergencies

#134 Ansible: Error in vault password

DESCRIPTION	Ansible cannot decrypt an Ansible Vault-encrypted variable or file.
ROOT CAUSE	Ansible Vault uses AES-256 encryption; the decryption requires the exact password used during encryption.
CAUSE	Wrong vault password provided, different vault ID used, or encrypted file corrupted.
SOLUTION	• Specify password file: ansible-playbook --vault-password-file=vault.pass • Or: ansible-playbook --ask-vault-pass • Re-encrypt with correct password: ansible-vault rekey vars.yml • Check ANSIBLE_VAULT_PASSWORD_FILE env var

#135 Helm: chart not found in repository

DESCRIPTION	Helm cannot locate the specified chart in the configured chart repository.
ROOT CAUSE	Helm searches local repository cache for the chart name and version; cache miss or wrong repo causes failure.
CAUSE	Repository not added, cache not updated, or chart name/version typo.
SOLUTION	• Add repo: helm repo add <name> <url> • Update cache: helm repo update • Search: helm search repo <chart> • Verify chart name and version spelling

#136 kubectl: error from server (Forbidden)

DESCRIPTION	A kubectl command fails because the requesting user or service account lacks permission.
ROOT CAUSE	Kubernetes RBAC evaluates the request against ClusterRoles/Roles bound to the principal; no matching allow rule results in denial.
CAUSE	Missing RBAC binding, wrong namespace, or cluster-admin rights not granted.
SOLUTION	• Check permissions: kubectl auth can-i <verb> <resource> --as=<user> • Create binding: kubectl create clusterrolebinding • Review existing bindings: kubectl get rolebinding,clusterrolebinding -A • Apply correct RBAC manifests

#137 GitHub Actions: Secret not found

DESCRIPTION	A workflow step fails because a referenced secret is not defined in the repository or organization secrets.
ROOT CAUSE	GitHub Actions injects secrets as environment variables only if they are defined at the appropriate scope; undefined secrets result in empty values.
CAUSE	Secret not added in Settings > Secrets, wrong name, or environment not configured.
SOLUTION	• Add secret: Settings > Secrets and variables > Actions • Verify secret name matches workflow reference exactly (case-sensitive) • Check environment secrets if using environments • Use ${{ secrets.SECRET_NAME }} syntax correctly

#138 SonarQube: Quality Gate failed

DESCRIPTION	A CI/CD pipeline build fails because SonarQube analysis found issues exceeding the Quality Gate thresholds.
ROOT CAUSE	SonarQube Quality Gates define conditions (coverage, bugs, code smells) that must pass; failing conditions block the build.
CAUSE	New code introduces bugs/vulnerabilities, coverage dropped, or duplications increased.
SOLUTION	• Review SonarQube project dashboard for specific failures • Fix reported issues in code • Adjust Quality Gate conditions if overly strict • Use sonar.qualitygate.wait=true to block appropriately • Add exclusions for test/generated code

#139 Packer: error waiting for SSH

DESCRIPTION	A Packer image build fails because it cannot SSH into the builder instance during provisioning.

ROOT CAUSE	Packer starts a VM/cloud instance and waits for SSH to become available to run provisioners; timeout means SSH never became reachable.
CAUSE	Security group/firewall blocking SSH, wrong SSH key, instance boot failure, or timeout too short.
SOLUTION	• Check cloud security group allows SSH from Packer host • Increase ssh_timeout in template • Verify ssh_username matches AMI user • Check cloud instance console for boot errors • Add debug logging: PACKER_LOG=1 packer build

#140 ArgoCD: Application out of sync

DESCRIPTION	ArgoCD reports an application as OutOfSync because the live cluster state does not match the desired Git state.
ROOT CAUSE	ArgoCD compares live Kubernetes resources with manifests in Git; any difference is flagged as out of sync.
CAUSE	Manual changes to cluster resources, image tag updated in Git, or drift from kubectl apply.
SOLUTION	• Review diff: argocd app diff <app> • Sync: argocd app sync <app> • Enable auto-sync: argocd app set <app> --sync-policy automated • Fix manual drift and let ArgoCD manage all changes through Git

Section 15: Performance & Observability (141–150)

Diagram: Linux Observability Tooling Map

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┌──────────────────────────────────────────────────────────────────┐
│                    LINUX PERFORMANCE TOOLS                       │
└──────────────────────────────────────────────────────────────────┘

Application  ──▶ strace (syscalls)  ──▶ ltrace (lib calls)
                 perf (CPU profile) ──▶ valgrind (memory)

CPU          ──▶ top / htop          ──▶ mpstat / sar
                 perf stat           ──▶ vmstat (context switches)

Memory       ──▶ free -h             ──▶ /proc/meminfo
                 smem                ──▶ valgrind memcheck

Disk I/O     ──▶ iostat -x           ──▶ iotop
                 blktrace            ──▶ smartctl (SMART)

Network      ──▶ ss / netstat        ──▶ tcpdump / wireshark
                 iftop / nethogs     ──▶ nmap / curl -v

Logs         ──▶ journalctl          ──▶ /var/log/*
                 dmesg               ──▶ audit.log (auditd)
```

#141 perf: Permission denied (kernel.perf_event_paranoid)

DESCRIPTION	The perf profiling tool is denied access to kernel performance counters.
ROOT CAUSE	The kernel parameter perf_event_paranoid controls access to perf events; high values restrict non-root profiling.
CAUSE	Default paranoid setting restricts perf to root only.
SOLUTION	• Temporarily allow: sysctl -w kernel.perf_event_paranoid=1 • Set permanently in /etc/sysctl.conf • Or run as root: sudo perf stat <command> • Use -1 to allow full access (security tradeoff)

#142 strace: attach: ptrace(PTRACE_SEIZE): Operation not permitted

DESCRIPTION	strace cannot attach to a running process to trace its system calls.
ROOT CAUSE	ptrace requires either root or matching UID AND the process must not have a no_ptrace LSM restriction.

CAUSE	Process owned by different user, Yama ptrace_scope set to 1+, or LSM restriction.
SOLUTION	• Run as root: sudo strace -p <PID> • Reduce Yama scope temporarily: sysctl kernel.yama.ptrace_scope=0 • Or use same UID as target process • Check AppArmor/SELinux ptrace restrictions

#143 Prometheus: many-to-many matching not allowed

DESCRIPTION	A PromQL query fails because a binary operation would result in a many-to-many label match.
ROOT CAUSE	PromQL binary operations require one-to-one or many-to-one matching; many-to-many is ambiguous and disallowed.
CAUSE	Missing group_left/group_right modifier, or wrong label set in on() clause.
SOLUTION	• Add group_left: metric1 * on(instance) group_left(job) metric2 • Specify matching labels explicitly with on() • Review cardinality of both sides of operation • Use without() to exclude non-matching labels

#144 Grafana: No data source found

DESCRIPTION	A Grafana dashboard panel cannot load because the configured data source is missing or unreachable.
ROOT CAUSE	Grafana queries data sources by name or UID; a missing or misconfigured data source returns no data or an error.
CAUSE	Data source deleted, Prometheus/InfluxDB unreachable, or dashboard imported without matching data source.
SOLUTION	• Add data source: Configuration > Data Sources • Verify data source URL is reachable from Grafana • Re-link panels to correct data source • Export/import dashboard with data source variables

#145 tcpdump: no suitable device found

DESCRIPTION	tcpdump cannot start because it cannot find or open the specified (or default) network interface.
ROOT CAUSE	tcpdump requires a capture-capable interface; if the name is wrong or libpcap cannot open it, capture fails.
CAUSE	Wrong interface name, interface down, insufficient permissions, or no interfaces available.
SOLUTION	• List interfaces: tcpdump -D • Specify correct interface: tcpdump -i eth0 • Run as root or with CAP_NET_RAW capability

	• Bring up interface: ip link set <iface> up

#146 valgrind: Mismatched free/delete/delete[]

DESCRIPTION	Valgrind reports a memory error where memory was allocated with one allocator but freed with another.
ROOT CAUSE	Valgrind's memcheck tracks allocations and their type; freeing with a mismatched function is undefined behavior.
CAUSE	Using free() on new[] allocated memory, or delete on malloc() allocated memory.
SOLUTION	• Match allocation and deallocation: malloc/free, new/delete, new[]/delete[] • Review code for mixed C/C++ allocation styles • Run valgrind with --leak-check=full for full report • Fix all mismatches to prevent heap corruption

#147 sar: Cannot open /var/log/sa/saXX

DESCRIPTION	The sar command cannot find historical data for the requested day.
ROOT CAUSE	sar reads from daily sa files written by the sadc collector; if collection was not running or data rotated, files are absent.
CAUSE	sysstat service not running, data not collected on that day, or rotation deleted files.
SOLUTION	• Enable collection: systemctl enable --now sysstat • Check service: systemctl status sysstat • View available files: ls /var/log/sa/ • Adjust HISTORY setting in /etc/sysstat/sysstat

#148 gdb: No symbol table loaded

DESCRIPTION	gdb cannot display function names or variable information because the binary lacks debug symbols.
ROOT CAUSE	Debug symbols (DWARF data) map binary addresses to source code; stripped binaries have no symbol table.
CAUSE	Binary compiled without -g flag, or strip run after compilation.
SOLUTION	• Recompile with debug symbols: gcc -g -O0 -o binary source.c • Or install debug package: apt install <package>-dbg • Use debuginfod: export DEBUGINFOD_URLS=https://debuginfod.elfutils.org/ • Load symbols manually: symbol-file /path/to/symbols

#149 iostat: device not found in /proc/diskstats

DESCRIPTION	iostat cannot report statistics for a specified device because it is not in the kernel's diskstats.

ROOT CAUSE	iostat reads /proc/diskstats which lists block devices the kernel knows about; unknown names produce no data.
CAUSE	Wrong device name, device not mounted, or device mapper name not used.
SOLUTION	• List devices: lsblk or cat /proc/diskstats • Use correct device name (e.g., dm-0 for LVM) • Verify device is accessible: ls /dev/<device> • For dm devices, map to name: dmsetup ls

#150 systemd-analyze: Permission denied

DESCRIPTION	systemd-analyze cannot read unit file details due to insufficient permissions.
ROOT CAUSE	Some systemd-analyze operations require reading protected unit files or kernel interfaces that are restricted to root.
CAUSE	Running as non-root user for a system-level analysis operation.
SOLUTION	• Run as root: sudo systemd-analyze blame • Or: sudo systemd-analyze critical-chain • For user units, no sudo needed: systemd-analyze --user blame • Install polkit to allow specific unprivileged analyze operations

DevOps Shack

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